

Function transformations

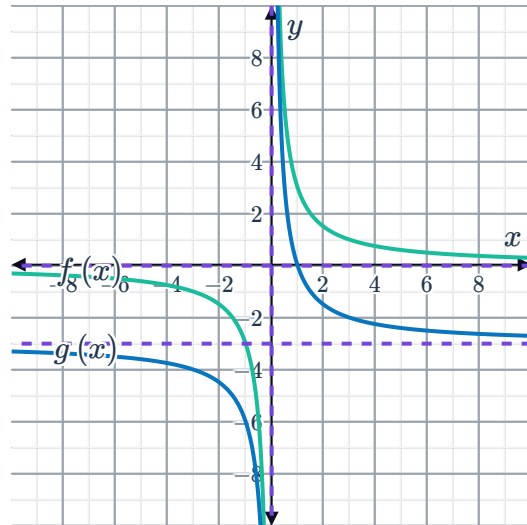


- 1 How should you translate the graph of $y = f(x)$ to get the graph of $y = f(x) + 4$?
- 2 How should you translate the graph of $y = g(x)$ to get the graph of $y = g(x + 6)$?
- 3 Describe the shift required to transform the graph of $y = a^x$ to get the graph of $y = a^{(x+5)}$.
- 4 For each of the following equations, determine what the new equation is when their graphs are moved as described:
 - a The graph of $y = 4x$ is moved up by 2 units.
 - b The graph of $y = x^3$ is moved to the right by 10 units.
 - c The graph of $y = 2^x$ is moved down by 9 units.
- 5 Suppose that the x -values of the x -intercepts of the graph of $y = f(x)$ are -5 and 6 . Find the corresponding x -values of the x -intercepts of the graph of:
 - a $y = f(x + 4)$
 - b $y = f(x - 4)$
 - c $y = 3f(x)$
 - d $y = f(-x)$
- 6 State the vertical stretch factor of:
 - a $y = 4x^3$
 - b $y = -4x^4$
 - c $y = 3x^5$
 - d $y = -5x^2$
- 7 Consider the function $f(x) = x^2 - 5$. Using function notation, describe the transformation of f that will result in the function:
 - a $g(x) = x^2 - 6$
 - b $h(x) = 16x^2 - 5$
 - c $k(x) = 3x^2 - 15$

8 Consider the graphs of $f(x) = \frac{3}{x}$ and $g(x)$:

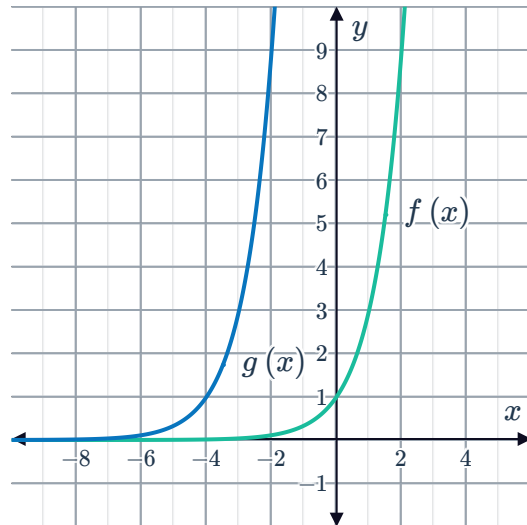


- a Write $g(x)$ in terms of $f(x)$.
- b State the equation of $g(x)$.



9 Consider the graphs of $f(x) = 3^x$ and $g(x)$:

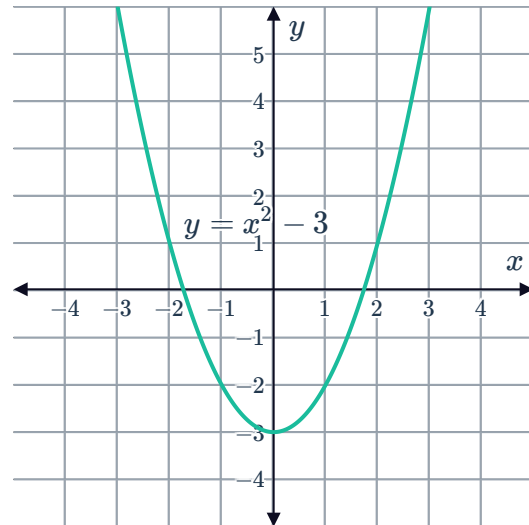
- a Write $g(x)$ in terms of $f(x)$.
- b State the equation of $g(x)$.



10 Consider the graph of $y = x^2 - 3$:

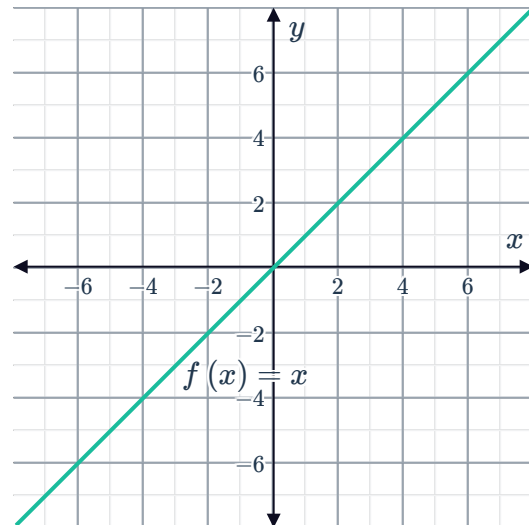


- a Write down the new equation when the graph of $y = x^2 - 3$ is moved upwards by 5 units.
- b Write down the new equation when the graph of $y = x^2 - 3$ is moved to the right by 3 units.



11 Consider the graph of $f(x) = x$.

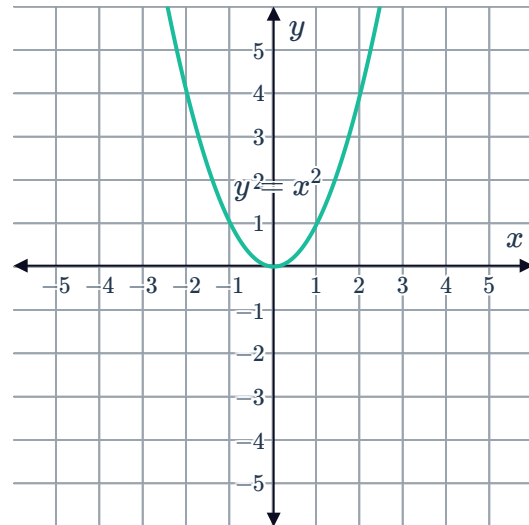
- a Write down the equation of the new function $g(x)$ which is formed by evaluating $f(x) + 4$.
- b Sketch the graph of $y = g(x)$.
- c Describe the transformation the graph of $f(x)$ undergone to become the graph of $g(x)$.



12 Consider the graph of $y = x^2$:

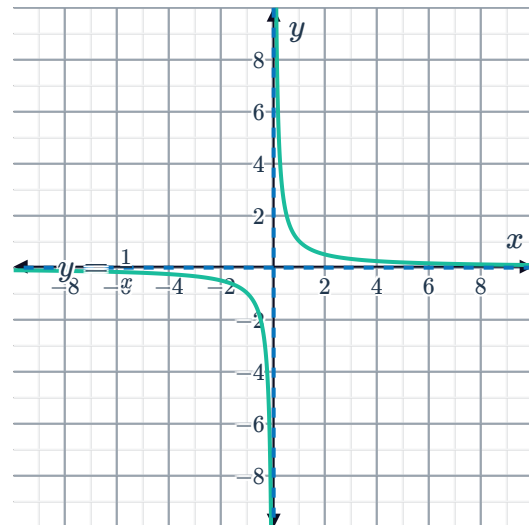


- a
- i Describe the transformation of the graph of $y = x^2$ that would obtain $y = x^2 - 3$.
 - ii Sketch the graph of $y = x^2 - 3$ on the same coordinate plane.
- b
- i Describe the transformation of the graph of $y = x^2$ that would obtain $y = (x - 2)^2$.
 - ii Sketch the graph of $y = (x - 2)^2$ on the same coordinate plane.



13 Consider the graph of $y = \frac{1}{x}$:

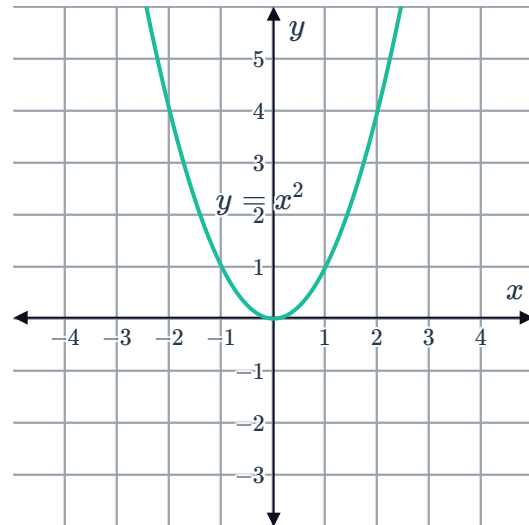
- a Describe the transformation of the graph of $y = \frac{1}{x}$ that would obtain $y = \frac{1}{x+2}$.
- b Sketch the graph of $y = \frac{1}{x+2}$ on the same coordinate plane.



14 Consider the graph of $y = x^2$:

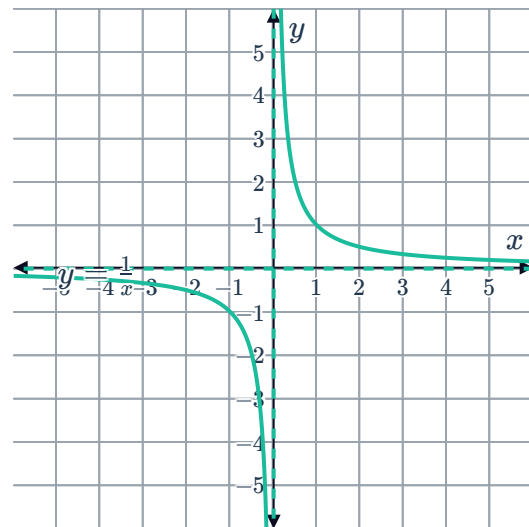


- a Describe the transformation of the graph of $y = x^2$ that would obtain $y = (x - 2)^2$.
- b Sketch the graph of $y = (x - 2)^2$.



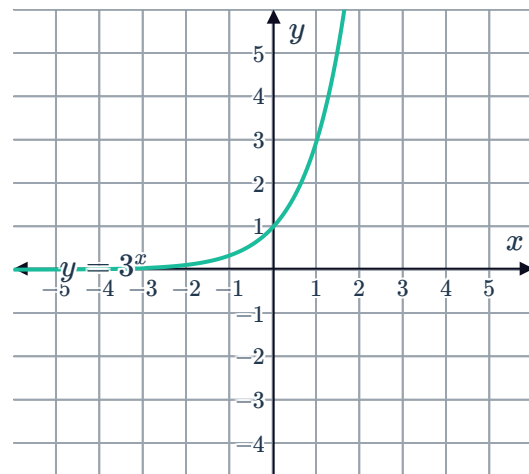
15 Consider the graph of $y = \frac{1}{x}$:

- a Describe the transformation of the graph of $y = \frac{1}{x}$ that would obtain $y = \frac{1}{x} + 3$.
- b Sketch the graph of $y = \frac{1}{x} + 3$.



16 Consider the graph of $y = 3^x$:

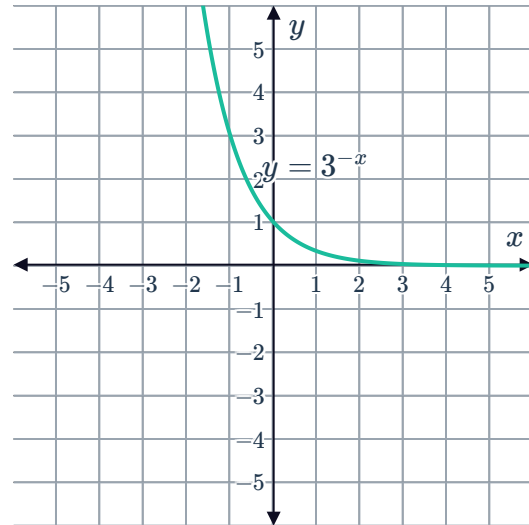
- a Describe the transformation of the graph of $y = 3^x$ that would obtain $y = 3^x - 4$.
- b Sketch the graph of $y = 3^x - 4$.



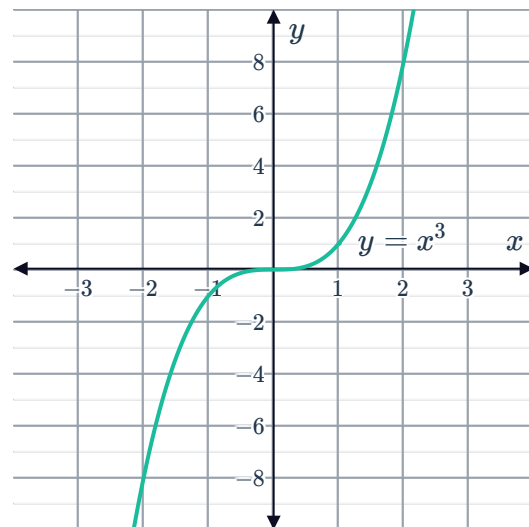


17 Consider the graph of $y = 3^{-x}$:

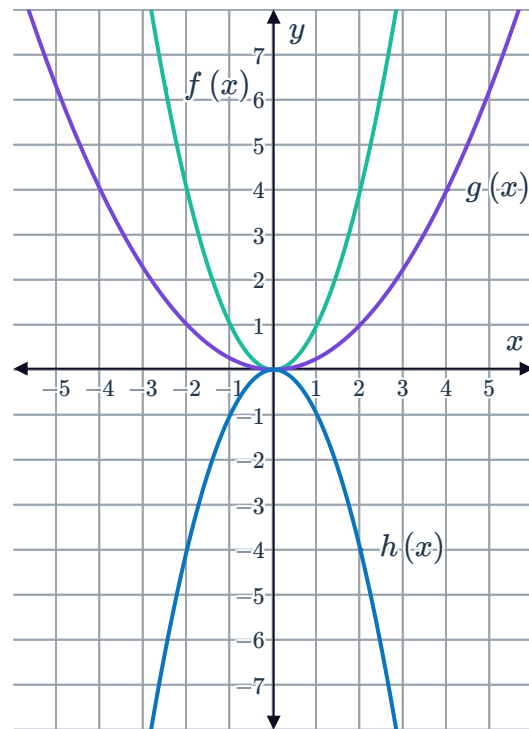
- a Describe the transformation of the graph of $y = 3^{-x}$ that would obtain $y = 3^{-x} + 2$.
- b Sketch the graph of $y = 3^{-x} + 2$.



18 Consider the graph of $y = x^3$:
Sketch the graph after it has undergone transformations resulting in the function $y = x^3 - 4$.



- 19 Three functions have been graphed on the coordinate plane. $g(x)$ and $h(x)$ are both transformations of $f(x)$.



- a State the equation of $f(x)$.
- b Write $g(x)$ in terms of $f(x)$ using the transformation shown in the graph.
- c State the equation of $g(x)$.
- d Write $h(x)$ in terms of $f(x)$ using the transformation shown in the graph.
- e State the equation of $h(x)$.

Multiple transformations

- 20 Describe the transformations that have occurred from $f(x)$ to $g(x)$:

- a From $f(x) = x^4$ to $g(x) = -5x^4 + 8$
- b From $f(x) = x^4$ to $g(x) = -7(x + 4)^4$
- c From $f(x) = x^2$ to $g(x) = -10(x + 8)^2 - 9$

- 21 The graph of $y = -x^2$ is moved to the right by 6 units and upwards by 5 units. What is the new equation of the graph?

- 22 Suppose that $(-4, 3)$ is a point on the graph of $y = g(x)$. Find the corresponding point on the graph of:

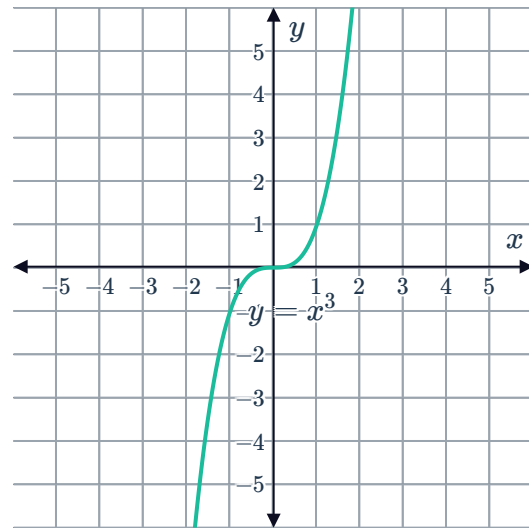
- a $y = g(x + 7) - 6$
- b $y = -6g(x - 4) + 6$
- c $y = g(6x + 1)$

- 23 Consider the function $f(x) = \sqrt{x}$. Write down the new function $g(x)$ which results from compressing $f(x)$ vertically by a factor of $\frac{1}{3}$, and compressing horizontally by a factor of $\frac{1}{2}$.

24 Consider the graph of $y = x^3$:

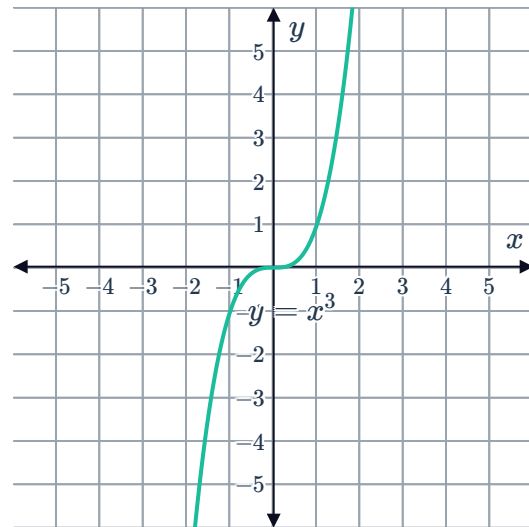


- a Describe the transformation of the graph of $y = x^3$ that would obtain $y = (x + 2)^3 + 4$.
- b Sketch the graph of $y = (x + 2)^3 + 4$.



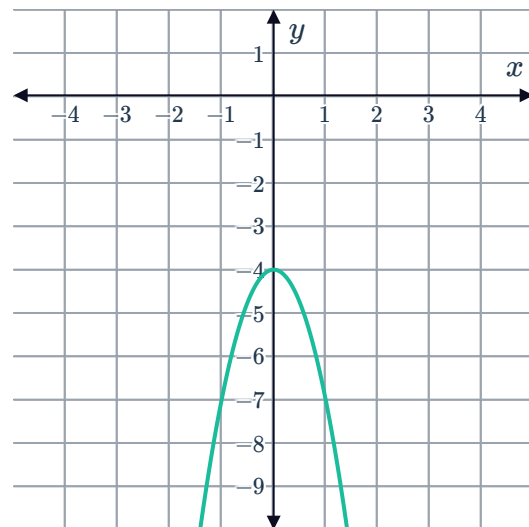
25 Consider the graph of $y = x^3$:

- a Describe the transformation of the graph of $y = x^3$ that would obtain $y = (x - 2)^3 - 3$.
- b Sketch the graph of $y = (x - 2)^3 - 3$.



26 Consider the following graph:

- a Describe the transformations of the graph of $y = x^2$ to the given graph.
- b Write down the equation of the given graph.

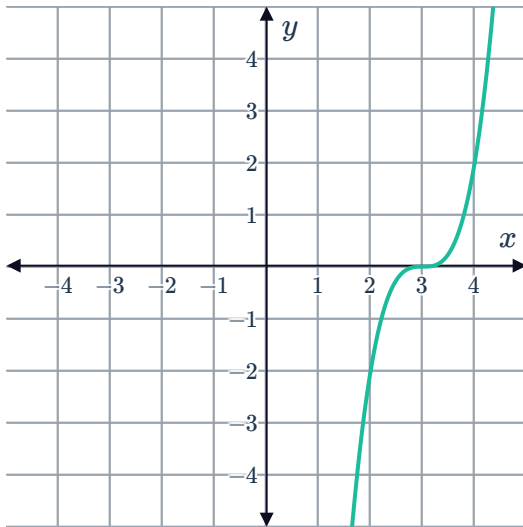


27 For each of the following graphs:

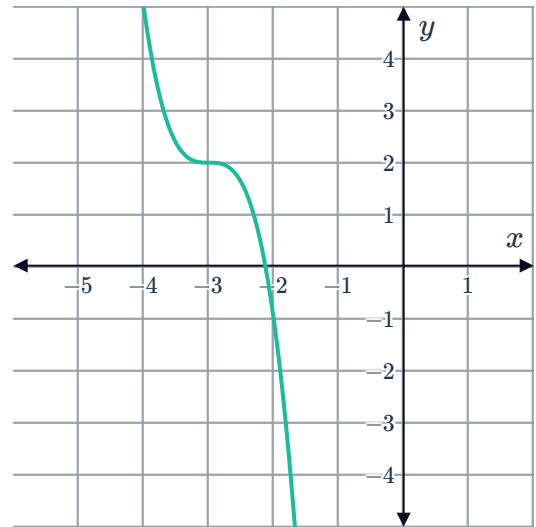


- i Describe the transformations of the graph of $y = x^3$ to the given graph.
- ii Write down the equation of the given graph.

a

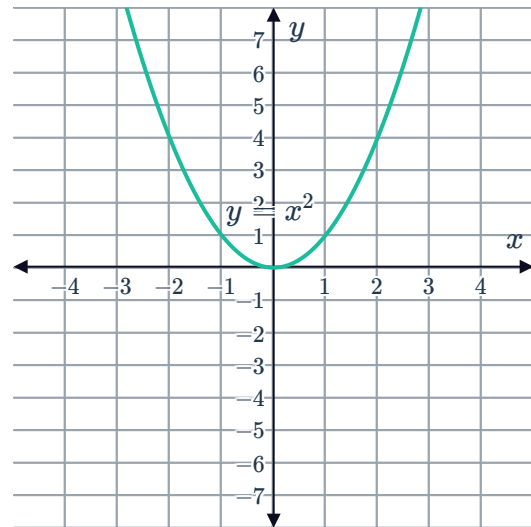


b



28 Consider the graph of $y = x^2$:

Sketch the graph after it has undergone transformations resulting in the function $y = 4x^2 - 2$.



29 Consider the function $f(x) = \sqrt{x}$.



- a Complete the table for $f(x)$:
- b Sketch a graph of the function $f(x)$.

x	0	1	4	9	16
$f(x)$					
$g(x)$					

- c Complete the table of values for the transformed function $g(x) = f(4x)$, giving all values in exact form.
- d Sketch a graph of $g(x)$.
- e Describe how $g(x)$ relates to the graph of $f(x)$.
- f Explain why the transformed function $h(x) = f(-4x)$ can be undefined for the values of x in the table.

30 The table below shows values that satisfy the function $f(x) = |x|$:

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3

- a Complete the table of values for each transformation of the function $f(x)$:

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$							
$h(x) = -2f(x)$							

- b Sketch the graphs of $f(x)$ and $g(x)$ on the same coordinate plane.
- c Describe the transformation undergone by $f(x)$ to become $g(x)$.
- d Sketch the graphs of $f(x)$ and $h(x)$ on the same coordinate plane.
- e Describe the transformation undergone by $f(x)$ to become $h(x)$.